

CURRENT WORK

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ABSTRACT. We briefly introduce the current work of the author, concerning about K-theory of some spectra constructed based on truncated Brown–Peterson spectra.

1. INTRODUCTION

Recall that for any $n \geq 0$ the truncated Brown-Peterson spectrum $\mathrm{BP}\langle n \rangle$ has an $\mathbb{E}_3$ -form for every $n \geq 0$, see [HW22]. Consider a pointed commutative monoid $M_e := \{0, 1, x, \dots, x^{e-1}\}$, we know that $\Sigma^\infty M_e$ is an $\mathbb{E}_\infty$ ring spectrum. So $\mathrm{BP}\langle n \rangle[x]/x^e := \mathrm{BP}\langle n \rangle \otimes \Sigma^\infty M_e$ can be equipped with an $\mathbb{E}_3$ -ring structure. Consider its p -completion, we arrive at an fp type n spectrum. If one view $\mathrm{BP}\langle n \rangle$ (p -adically) as (chromatic) higher version of $\mathbb{Z}_p$, then one can consider $\mathbb{Z}_p[x]/x^e$ or $\mathbb{Z}_p/p^e$. In other words, one can also consider (the p -completion of) $\mathrm{BP}\langle n \rangle/v_n^e$. According to a result in [Bur22], $\mathrm{BP}\langle n \rangle/v_n^e$ can be equipped with an $\mathbb{E}_2$ ring structure for $e > 2$. And $(\mathrm{BP}\langle n \rangle/v_n^e)_p^\wedge$ is an fp type $n - 1$ spectrum.

According to Rognes' redshift conjecture, we then have following questions.

Question 1.1. Is $K((\mathrm{BP}\langle n \rangle[x]/x^e)_p^\wedge)_p^\wedge$ an fp spectrum of type $n + 1$?

And also, multi variables version.

Question 1.2. Is $K((\mathrm{BP}\langle n \rangle[x_1, \dots, x_m]/(x_1^{e_1}, x_2^{e_2}, \dots, x_n^{e_n}))_p^\wedge)_p^\wedge$ an fp spectrum of type $n + 1$?

Question 1.3. For $e > 2$, is $K((\mathrm{BP}\langle n \rangle/v_n^e)_p^\wedge)_p^\wedge$ an fp spectrum of type n ?

Using the trace method and techniques in [HW22], we can actually reduce to following questions.

Question 1.4. Does $\mathrm{THH}(\mathrm{BP}\langle n \rangle[x]/x^e)_p^\wedge$ satisfy the Segal conjecture (the cyclotomic Frobenius map induce isomorphism on $V(n + 1)$ -homology for sufficient high degrees)?

Question 1.5. Does $\mathrm{THH}(\mathrm{BP}\langle n \rangle/v_n^e)_p^\wedge$ satisfy the Segal conjecture (the cyclotomic Frobenius map induce isomorphism on $V(n)$ -homology for sufficient high degrees)?

And for Question 1.4 the answer turns out to be negative, the rough idea is $\mathrm{THH}(\Sigma^\infty M_e)$ is not bounded above (according to results in [HM97].) For Question 1.5, if one want to use same trick, one need to compute the Adams E_2 page of $\mathrm{BP}\langle n \rangle/v_n^e$ (ongoing).

But one could still ask for some computational results, for instance, the following questions.

Since the $V(1)$ -homology of $\mathrm{TC}(\mathrm{BP}\langle 1 \rangle)$ is computed, and also $V(1) \otimes \mathrm{THH}(\mathrm{BP}\langle 1 \rangle; \mathbb{Z}_p)^{C_{p^k}}$ [AR02], together with the decomposition of $\mathrm{THH}(\Sigma^\infty M_e)$ one can try to compute $V(1) \otimes \mathrm{TC}(\mathrm{BP}\langle 1 \rangle[x]/x^e; \mathbb{Z}_p)$ using those results.

And also one can consider the relative THH . Since we have an $\mathbb{E}_\infty$ -map $\mathrm{MU}[x] \rightarrow \mathrm{MU}[x]/x^e$, thus we get an $\mathbb{E}_3$ - $\mathrm{MU}[x]$ -algebra form of $\mathrm{BP}\langle n \rangle[x]/x^e$.

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And also, we can use the strategy of [HW22, §2] to construct an at least an $\mathbb{E}_1$ -MU[x]-algebra. And we have following commutative diagram.

$$\begin{array}{ccccc}
\mathrm{BP}\langle n \rangle \in \mathrm{Alg}_{\mathrm{MU}}^{(1)} & \longleftarrow & \mathrm{Alg}_{\mathrm{MU}}^{(1)} \ni \mathrm{BP}\langle n \rangle & \longleftarrow & \mathrm{Alg}_{\mathrm{MU}}^{(3)} \ni \mathrm{BP}\langle n \rangle \\
\downarrow & & \downarrow & & \downarrow \\
\mathrm{BP}\langle n \rangle[x] \in \mathrm{Alg}_{\mathrm{MU}[x]}^{(1)} & & \mathrm{Alg}_{\mathrm{MU}[x]/x^e}^{(1)} \ni \mathrm{BP}\langle n \rangle[x]/x^e & \longleftarrow & \mathrm{Alg}_{\mathrm{MU}[x]/x^e}^{(3)} \ni \mathrm{BP}\langle n \rangle[x]/x^e \\
\downarrow & & \downarrow & & \downarrow \\
\mathrm{BP}\langle n \rangle[x]/x^e \in \mathrm{Alg}_{\mathrm{MU}[x]}^{(1)} & \longleftarrow & \mathrm{Alg}_{\mathrm{MU}[x]/x^e}^{(1)} \ni \mathrm{BP}\langle n \rangle[x]/x^e & & \mathrm{Alg}_{\mathrm{MU}[x]}^{(3)} \ni \mathrm{BP}\langle n \rangle[x]/x^e \\
& \searrow & \downarrow & \swarrow & \\
& & \mathrm{BP}\langle n \rangle[x]/x^e \in \mathrm{Alg} & &
\end{array}$$

Therefore one could try to compute the following things.

Question 1.6. Compute $V(1) \otimes \mathrm{TC}(\mathrm{BP}\langle 1 \rangle[x]/x^e; \mathbb{Z}_p)$.

Question 1.7. Compute

$$\mathrm{THH}((\mathrm{BP}\langle n \rangle[x]/x^e)/\mathrm{MU}[x]).$$

Question 1.8. Compute

$$\mathrm{THH}((\mathrm{BP}\langle n \rangle/v_n^e)/\mathrm{MU}).$$

2. SOME RESULTS

According to some ideas and techniques presented before, we have some results.

Theorem A. *There are $\mathbb{E}_3$ -algebras $\mathrm{BP}\langle n \rangle[x]/x^e, \Lambda_{\mathrm{BP}\langle n \rangle}(x), \mathrm{BP}\langle n \rangle/v_n^e$. Any of them is a finite $\mathrm{BP}\langle n \rangle$ -module, thus we can view them as finite algebras over $\mathrm{BP}\langle n \rangle$ (the ring of integers of a height n local field).*

Theorem B 1. *The ring spectra $\mathrm{THH}(\mathrm{BP}\langle n \rangle[x]/x^e/\mathrm{MU}[x]), \mathrm{THH}(\mathrm{BP}\langle n \rangle/v_n^e/\mathrm{MU})$ are even.*

Theorem C. *We have*

$$\begin{aligned}
L_{K(n+1)}(K(\mathrm{BP}\langle n \rangle[x]/x^e)) &\not\cong 0, \\
L_{K(n+1)}(K(\Lambda_{\mathrm{BP}\langle n \rangle}(x))) &\not\cong 0, \\
L_{K(n)}(K(\mathrm{BP}\langle n \rangle/v_n^e)) &\not\cong 0.
\end{aligned}$$

Moreover, we learn that v_{n+1} is non-nilpotent in $\mathrm{TC}^-(\mathrm{BP}\langle n \rangle[x]/x^e/\mathrm{MU}[x])$ and $\mathrm{TC}^-(\Lambda_{\mathrm{BP}\langle n \rangle}(x)/\Lambda_{\mathrm{MU}}(x))$. And v_n is non-nilpotent in $\mathrm{TC}^-(\mathrm{BP}\langle n \rangle/v_n^e/\mathrm{MU})$.

Theorem D. *Let F be any type $n+1$ finite complex, then the Frobenius*

$$F_*(\mathrm{THH}(\mathrm{BP}\langle n \rangle[x])) \rightarrow F_*(\mathrm{THH}(\mathrm{BP}\langle n \rangle[x])^{tC_p})$$

induces isomorphism on large degrees.

Theorem E. *The localizations*

$$\begin{aligned}\mathrm{TC}(\mathrm{BP}\langle 1\rangle[x])_p^\wedge &\rightarrow L_2^f(\mathrm{TC}(\mathrm{BP}\langle 1\rangle[x])_p^\wedge) \\ \mathrm{K}(\mathrm{BP}\langle 1\rangle[x]_p^\wedge)_p^\wedge &\rightarrow L_2^f(\mathrm{K}(\mathrm{BP}\langle 1\rangle[x]_p^\wedge)_p^\wedge)\end{aligned}$$

all have bounded above fibers.

Theorem F. *Let F be any type $n + 2$ finite complex, then $F \otimes \mathrm{TR}(\mathrm{BP}\langle n\rangle[x]/x^e)$ is not bounded, where $e > 1$. In otherwords, the Lichtenbaum–Quillen property fails for $\mathrm{BP}\langle n\rangle[x]/x^e$, when $e > 1$.*

In conclude, by Theorem E, one could say the the $\mathbb{A}^1$ -invariance still exists even for ring spectra. By Theorem F, one could say K-theory still sensitive for singularities, even for ring spectra.

REFERENCES

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